

Floating Point

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Floating Point Representation

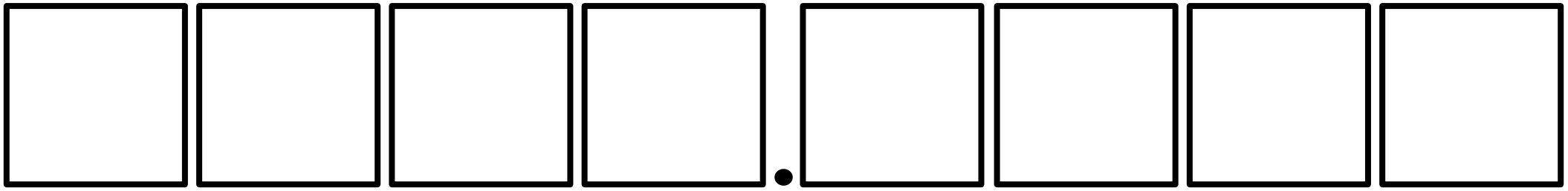
- What is **floating point representation**?
  - A way to represent very large and very small numbers efficiently
- A standard called **IEEE-754** defines how modern systems handle these values.
- We will start with simpler models (**microfloat** and **minifloat**) to build up to **IEEE-754.**

Motivation

- Learning floating point isn’t just about learning how numbers are stored
  - It teaches you how computers approximate reality,
  - Why rounding errors occur,
  - And why certain calculations don’t behave as expected.
- Floating point teaches you how to think like a computer scientist
  - You learn about **design tradeoffs**
  - And how to balance precision, range, performance, and practicality.

Fixed-Point Representation

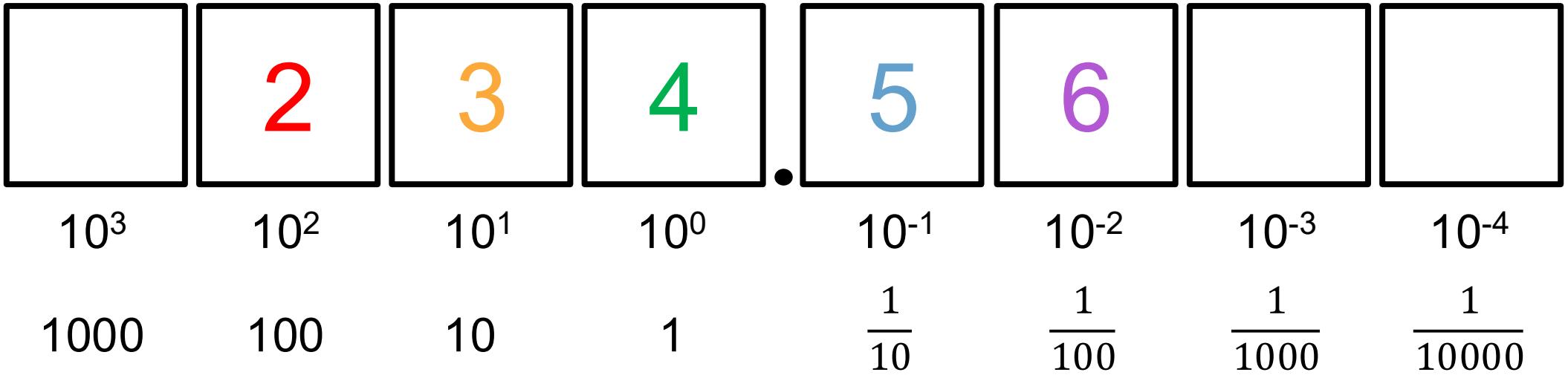
Place Values in Base-10



10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}
1000	100	10	1	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10000}$

Example: 234.56

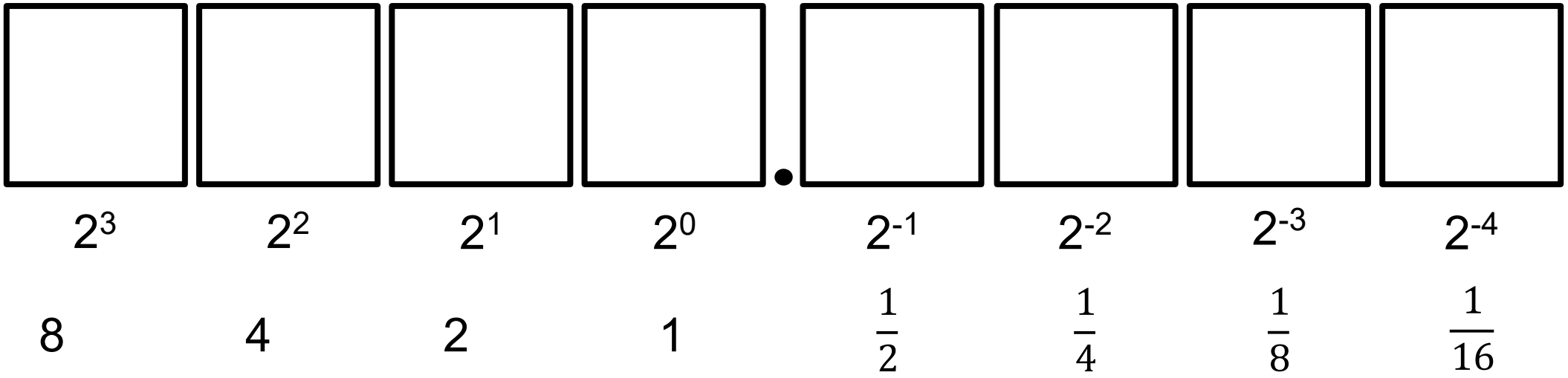
Place Values in Base-10



Example: 234.56

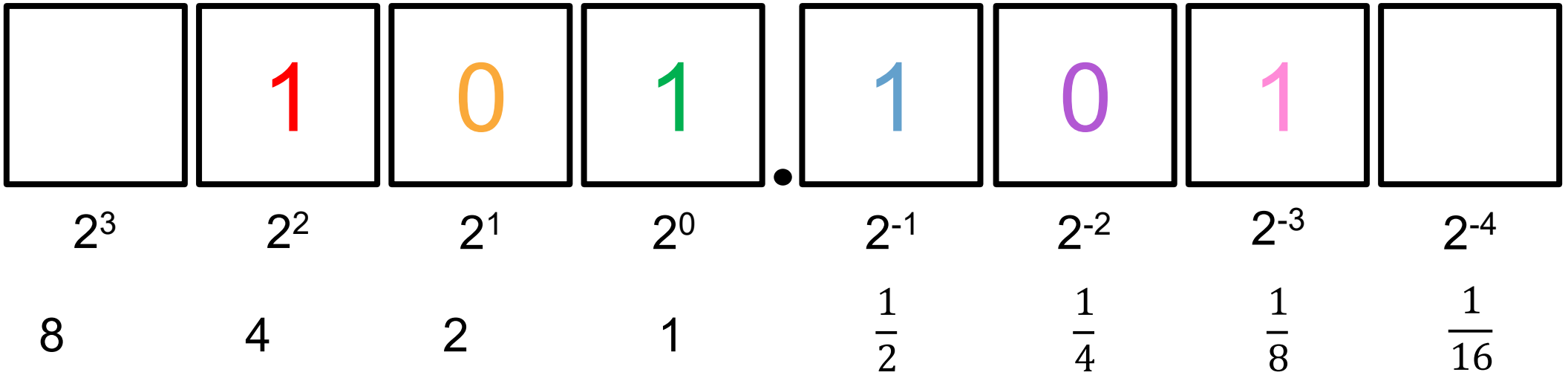
(2x100) + (3x10) + (4x1) + (5x0.5) + (6x0.01)

Fixed Point Representation (Unsigned)



Example: 101.101

Fixed Point Representation (Unsigned)



Example: 101.101

(1x4) + (0x2) + (1x1) + (1x0.5) + (0x0.25) + (1x0.125)

5.625

Conversions

- What is 01110.1011_2 in decimal?

- What is 12.75 in unsigned fixed-point binary representation?

Conversions

- What is 01110.1011_2 in decimal?
  - 14 + 0.5 + 0.125 + 0.0625 = 14.6875

- What is 12.75 in unsigned fixed-point binary representation?
  - 12 = 0b01100
  - 0.75 = 0.5 + 0.25
  - 12.75 = 0b1100.11

Scientific Notation

- Scientific notation is used to represent large or small numbers more compactly by reducing the number of digits displayed.

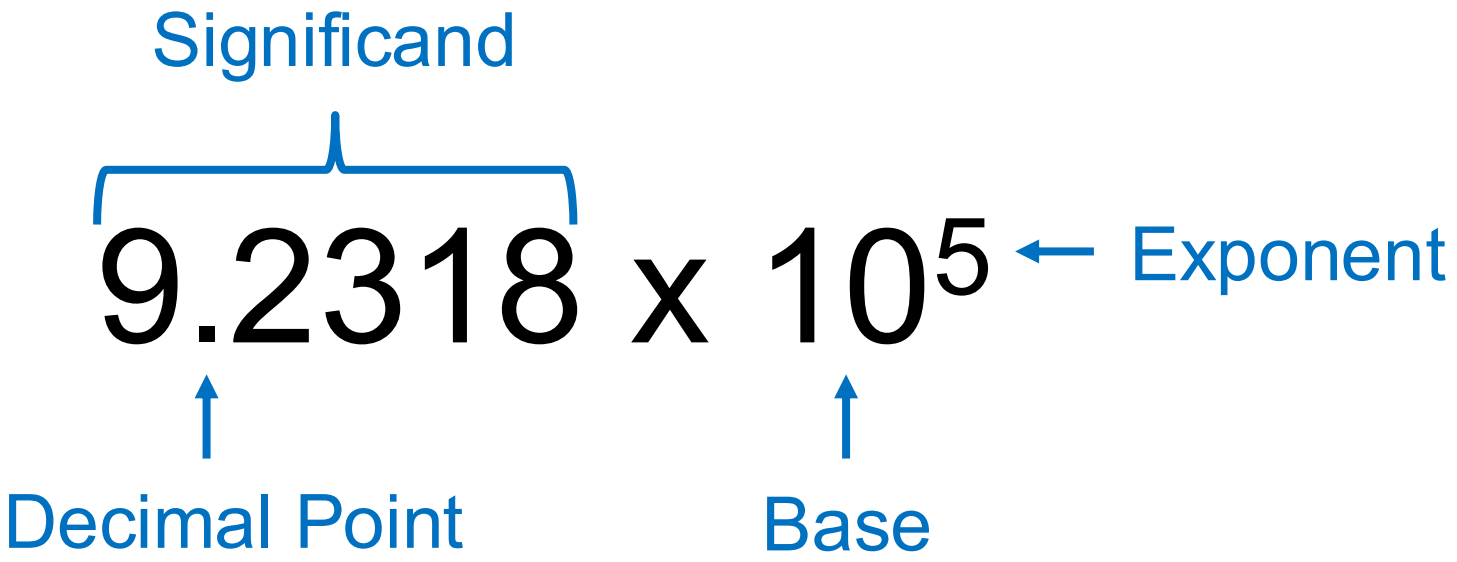
3.0×10^{8} m/s

6.022×10^{23} mol-1

Scientific Notation Terms

$$9.2318 \times 10^{5}$$

Scientific Notation Terms

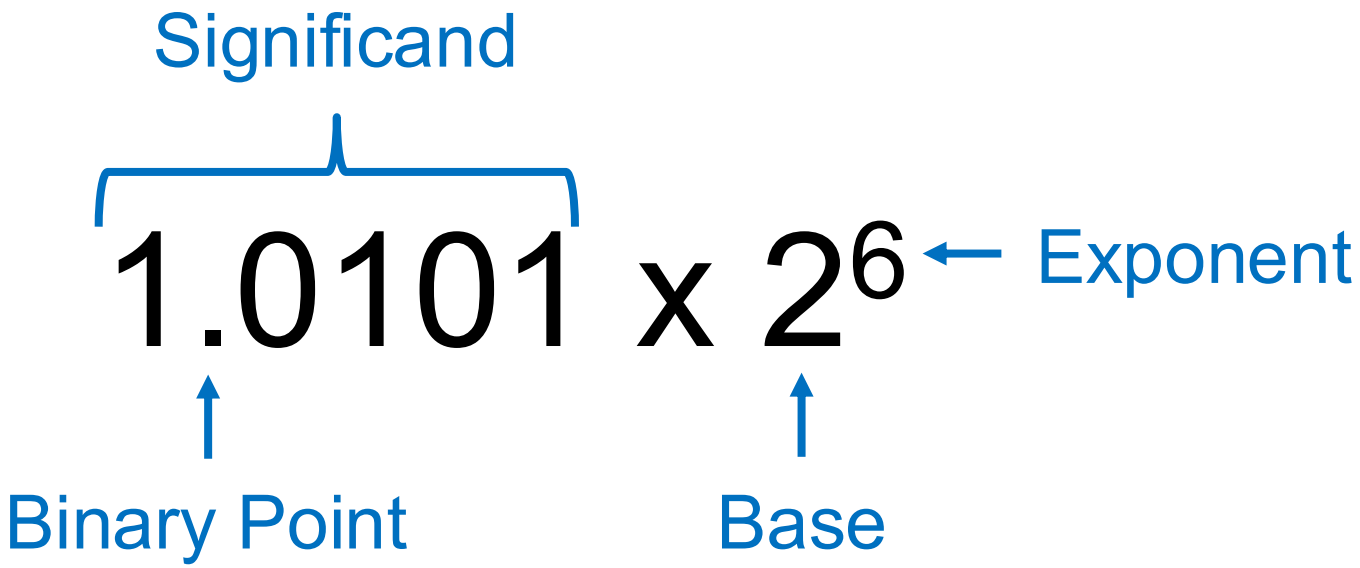


Normalized Form: 1 <= |significand| < 10

Binary in Normalized Form

$$1.0101 \times 2^{6}$$

Binary in Normalized Form



Normalized Form: 1 <= |significand| < 2

Conversions

- What is $1.0101_2 \times 2^6$ in decimal?

- What is 10.5 in normalized binary form?

Conversions

- What is $1.0101_2 \times 2^6$ in decimal?
  - $1.0101_2 \times 2^6 = 1010100 = 84$

- What is 10.5 in normalized binary form?
  - $1.0101_2 \times 2^3$

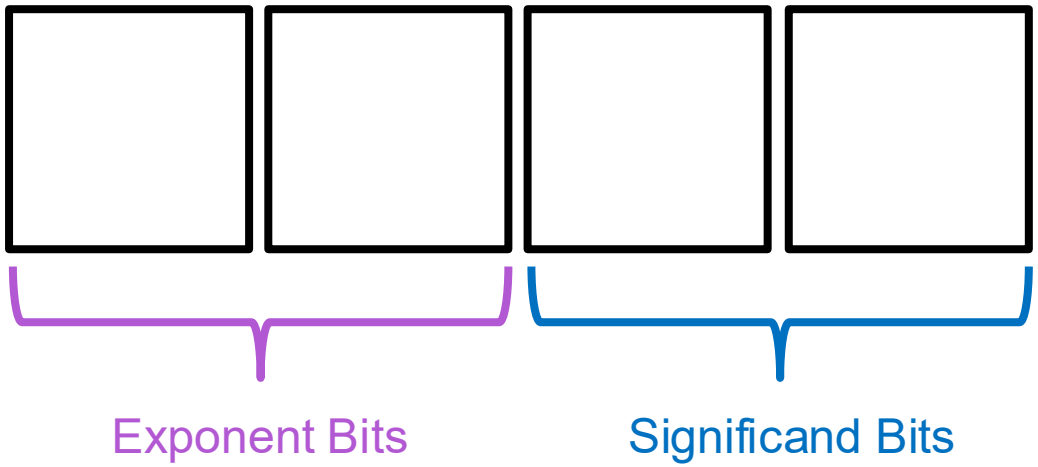
Microfloat

Floating Point

- The full value is not stored directly
  - (Unlike the **fixed-point** notation we saw earlier)
- Instead, we store
  - **Significand** (without the leading one)
  - **Exponent**
- This design allows for a much larger **range** of values
  - But it in turn trades off **precision**

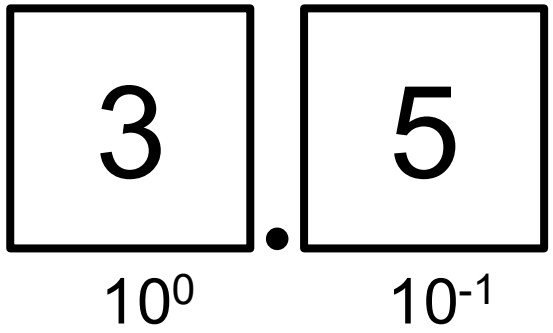
Microfloat

- We'll start by looking at a simple design called **microfloat**.
- Let's say we have 4 bits to represent our fractional value. Let's use two bits for the exponent and two bits for the significand.



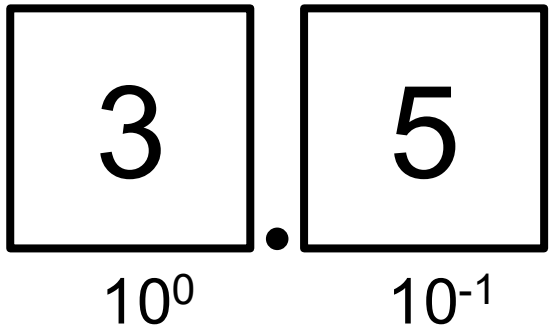
Encoding Fractional Values

Decimal Place-Value

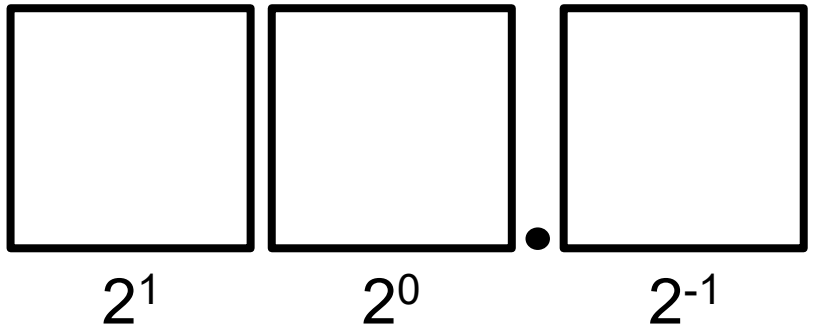


Encoding Fractional Values

Decimal Place-Value

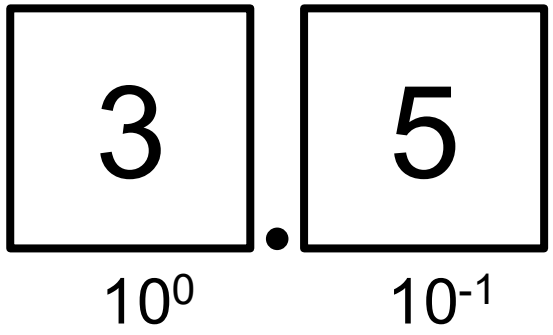


Binary Place-Value

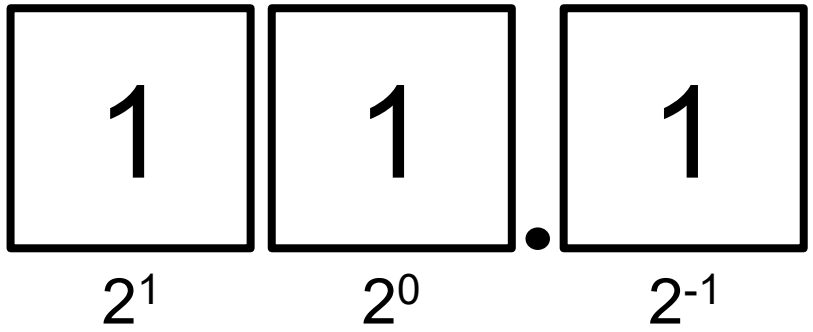


Encoding Fractional Values

Decimal Place-Value

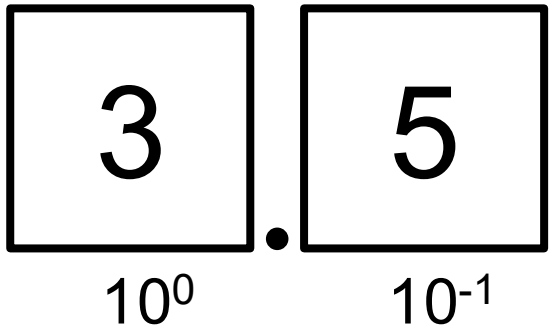


Binary Place-Value

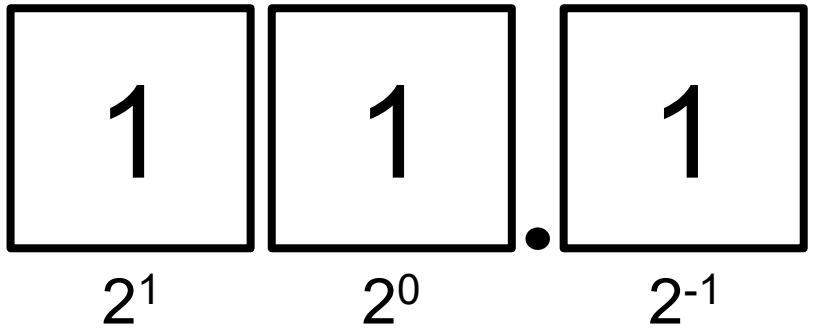


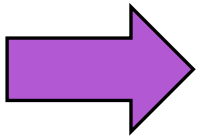
Encoding Fractional Values

Decimal Place-Value



Binary Place-Value





1.11×2^1

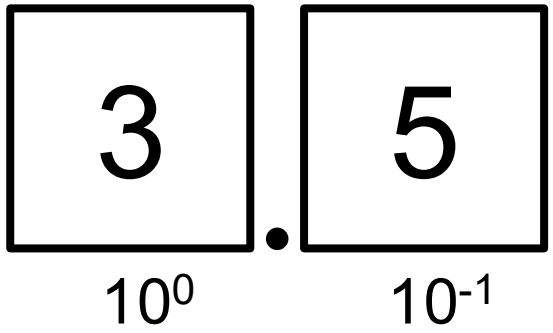
Exponent

Significand

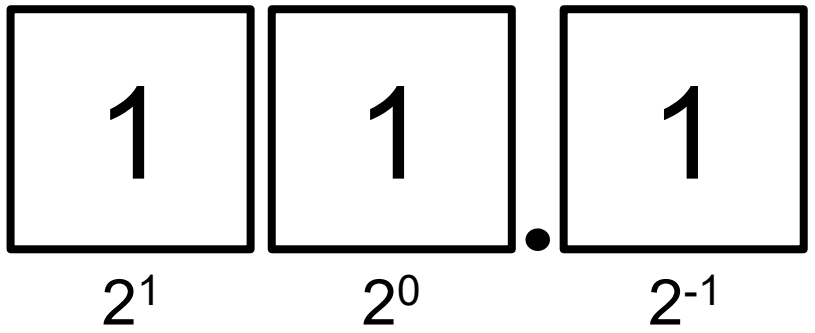
The leading 1 is not stored!

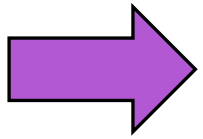
Encoding Fractional Values

Decimal Place-Value



Binary Place-Value





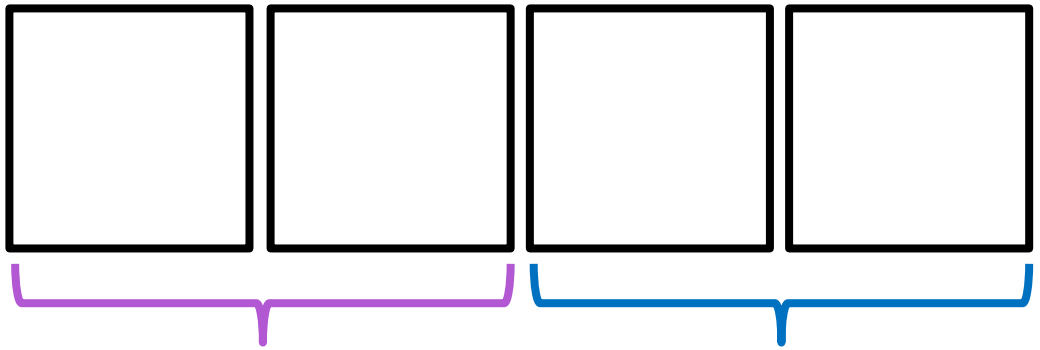
1.11×2^1

Exponent

Significand

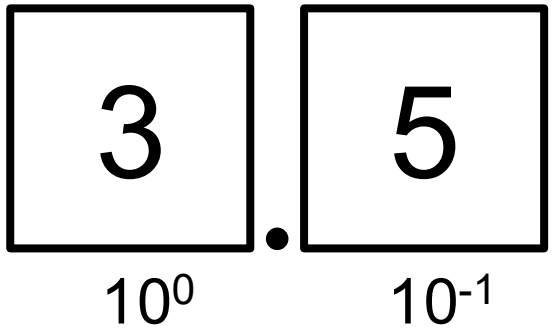
The leading 1 is not stored!

Microfloat

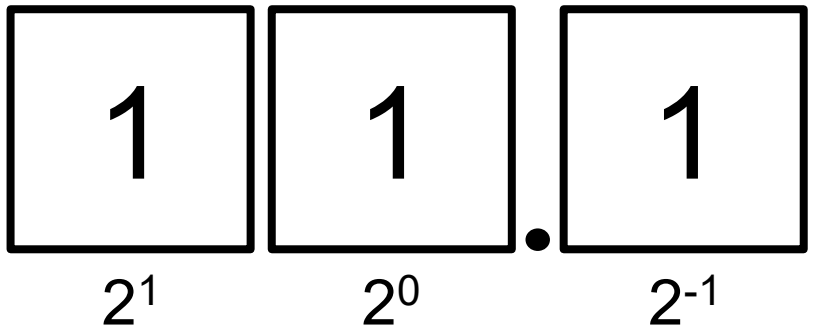


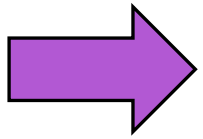
Encoding Fractional Values

Decimal Place-Value



Binary Place-Value



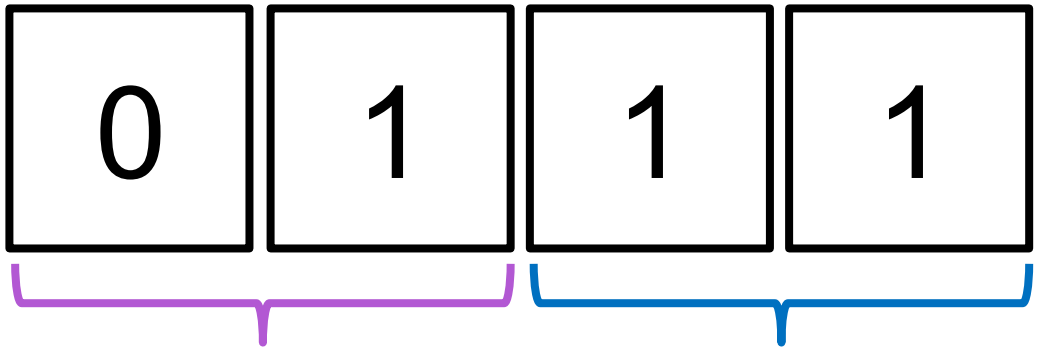


1.11×2^1

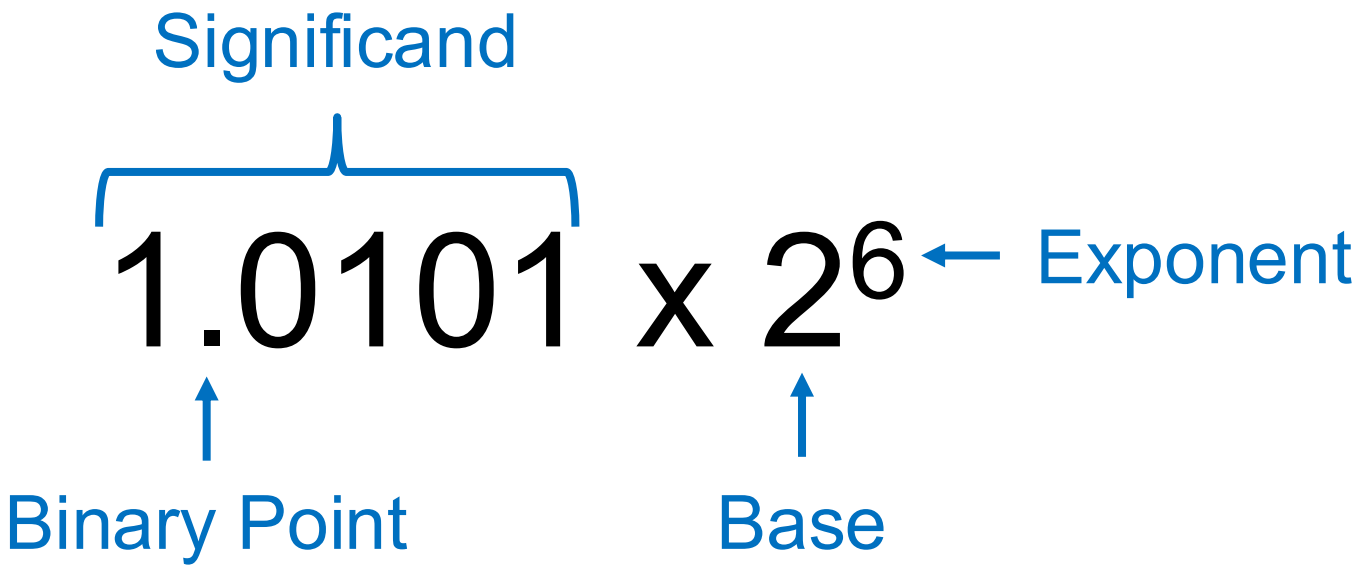
Exponent

Significand
The leading 1 is not stored!

Microfloat



Binary in Normalized Form



In normalized form, 1 <= |significand| < 2

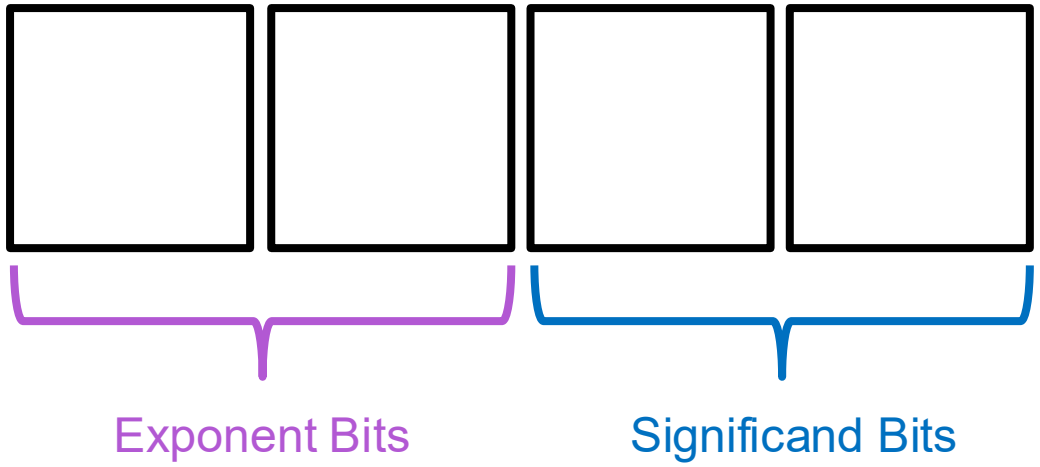
This means the value to the left of the binary point is always 1.
We can increase the precision by not storing this 1!

Implicit 1

- In normalized binary floating point, every (nonzero) number has a leading 1 before the binary point.
- Since that leading one is always there, we don’t need to store it
  - Hence the name “Implicit 1”
- By not storing it, we free up one extra bit for the significand
  - This extra bit increase **precision**
- The **range** is unaffected by this
  - (Since the range is determined by the exponent bits)

Example: Decimal to Microfloat V1

7



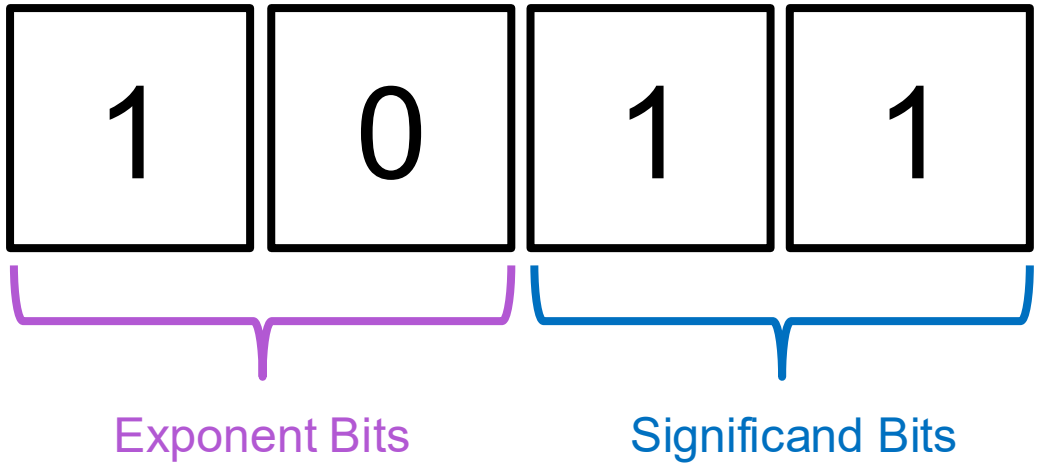
Example: Decimal to Microfloat V1

7

0b111

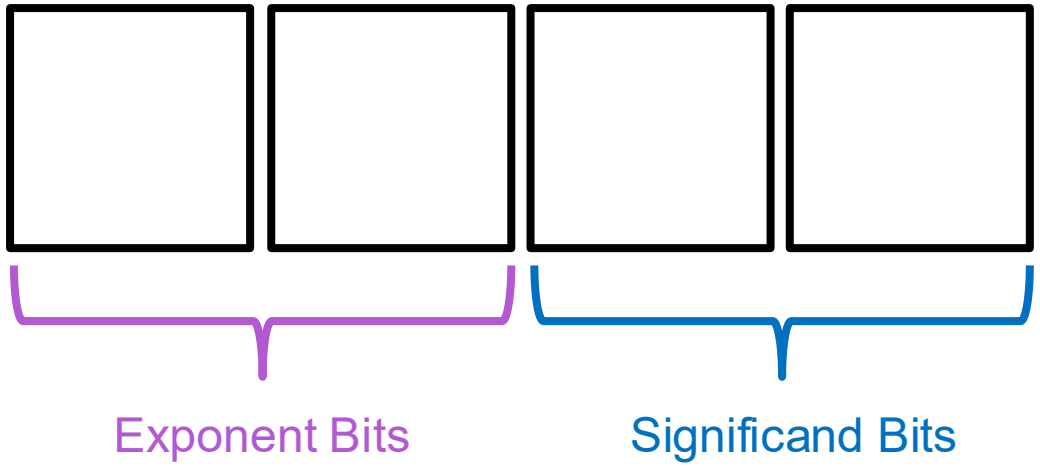
$1.11_2 \times 2^2$

1.<significand bits> x 2<exponent bits>



Example: Decimal to Microfloat V1

1.25



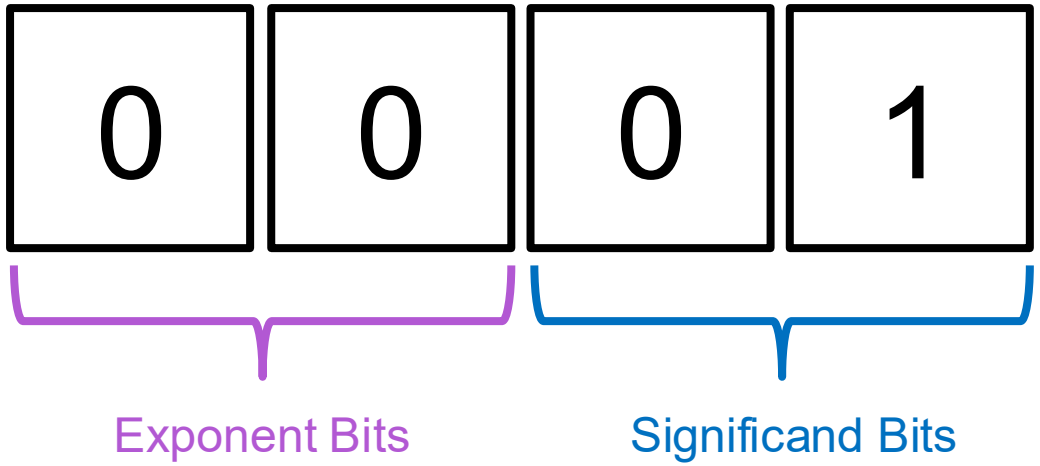
Example: Decimal to Microfloat V1

1.25

0b1.01

$1.01_2 \times 2^2$

1.<significand bits> x 2<exponent bits>



Example: Microfloat V1 to Decimal



Example: Microfloat V1 to Decimal



1.<significand bits> x $2^{<\text{exponent bits}>}$

1.01 x 2^{1}

2.5

Example: Microfloat V1 to Decimal



Example: Microfloat V1 to Decimal



1.<significand bits> x $2^{<\text{exponent bits}>}$

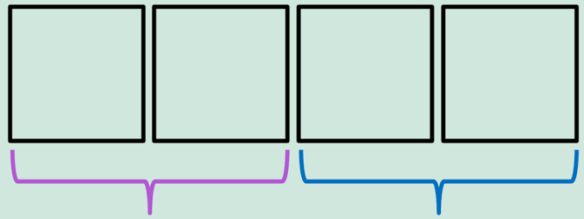
1.01 x 2^{3}

10

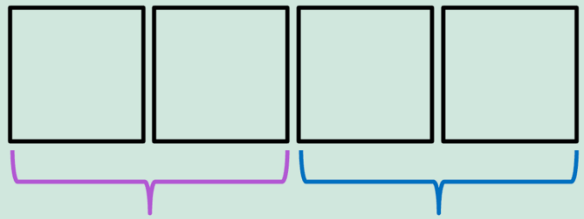
Microfloat V1

- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V1 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) 1.75
- B) 2.75
- C) 5
- D) 12
- E) 16

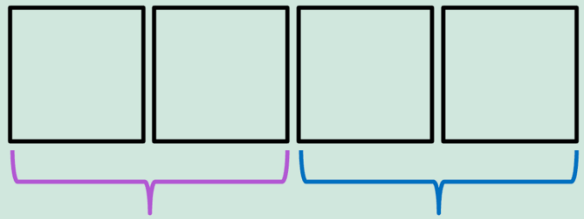
A



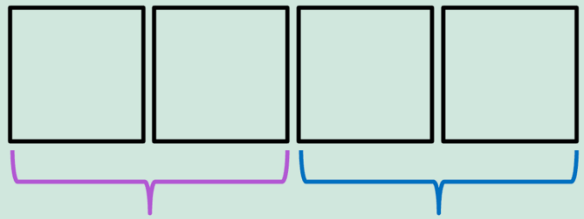
B



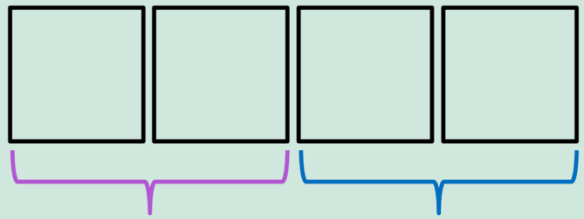
C



D



E



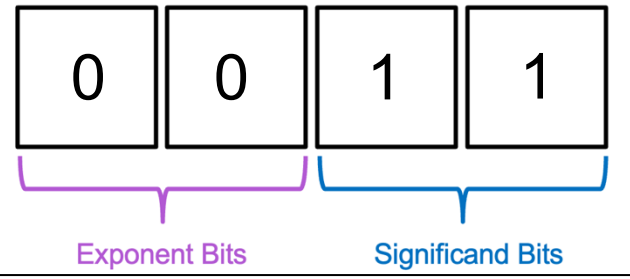
Microfloat V1

- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V1 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) $1.75 = 1.11_2 \times 2^0 \Rightarrow 0011_2$
- B) $2.75 = 1.011_2 \times 2^1$

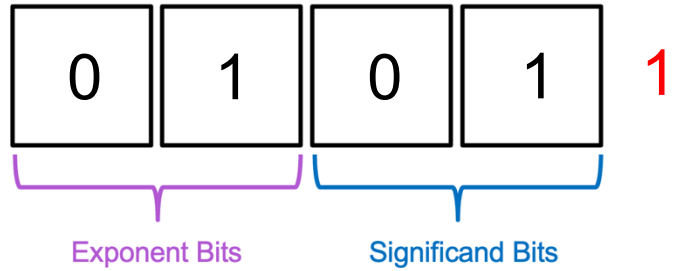
  $\Rightarrow$ not enough precision
- C) $5 = 1.01_2 \times 2^2 \Rightarrow 1001_2$
- D) $12 = 1.100_2 \times 2^3 \Rightarrow 1110_2$
- E) $16 = 1_2 \times 2^4$

  $\Rightarrow$ not enough range

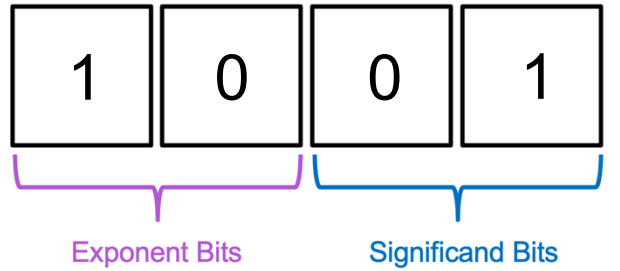
A



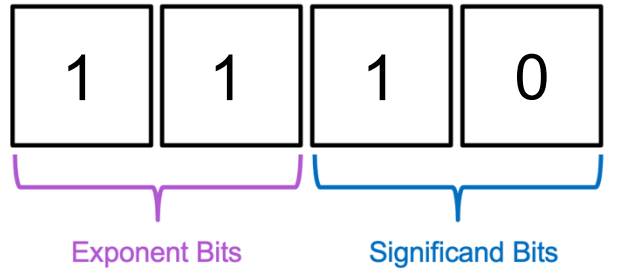
B



C



D



E

